

Problem: The remission survival times in week of 19 patients are as follows:

76, 166, 110, 144, 26, 118, 131, 14, 49, 153, 66, 36, 32, 11, 11, 15, 75, 85, 100⁺
("+" stands for censoring).

- (i) Estimate the survival probabilities using the Kaplan-Meier method.
- (ii) Estimate the standard error of survival probability for each time point.
- (iii) Construct the survival curve with its 95% confidence belts.
- (iv) Comment on your findings.

Solution: (i) and (ii): Estimating the survival probabilities and standard error of survival probability using the Kaplan-Meier method:

R-code and output:

```
time = c(75, 166, 110, 144, 26, 118, 131, 14, 49, 153, 66, 36, 32,  
         11, 11, 15, 75, 85, 100)
```

```
status = c(rep(1, 17), rep(0, 2))
```

```
dataset = data.frame(time, status)
```

```
dataset
```

```
#(i) and (ii)
```

```
library(survival)
```

```
km.fit = survfit(Surv(time, status == 1) ~ 1, data = dataset)
summary(km.fit)
```

Plot:

#(iii)

```
plot(km.fit, col = 'red', lwd = 2, main = 'Plot of estimated
survival curve with 95 % confidence belts', col.main =
'blue', xlab = 'Survival Time', col.lab = 'blue', xlim = c(0, 100),
ylab = 'Survival Probability', col.lab = 'blue', conf.int = 0.95)
```

```
abline(h = 0.75, v = 26, col = 4, lwd = 2)
```

Plot:

```
time = km.fit$time [-17]
```

```
s = km.fit$surv [-17]
```

```
km.fit$time
```

```
time
```

```
s
```

```
x = log(time)
```

```
y = log(-log(s))
```

```
plot(x, y, xlim = c(1.5, 5), ylim = c(-3, 2), main = "Weibull plot  
of remission data", xlab = "log(t)", ylab = "log(-log(s))")
```

```
abline(lsfitted(x, y), col = 4, lwd = 2)
```

```
abline(h = 0, col = 3, lwd = 2)
```

```
abline(h = c(0, -3.06), col = c(3, 2), lwd = 2)
```

```
abline(v = 4.48, col = 4, lwd = 2)
```

Plot:

(iv) Comment on my findings:

The Kaplan-Meier estimator is used to estimate the survival probabilities and their standard errors at different event times. The estimated survival probability decreased progressively as time increased, indicating that the likelihood of remaining in remission declined over time. On the other hand, the standard error increased gradually over time, indicating greater uncertainty in the survival estimates at later time points. From the table, at time 26, there is 15 individuals at risk and 1 event occurred. The Kaplan-Meier estimate of the survival probability is 0.7368, indicating that approximately 73.68% of the subjects are expected to survive beyond time 26. The 95% confidence interval for the true survival probability ranged

from 0.5632 to 0.964, suggesting that the survival probability at time 26 is likely to lie within its interval.

The survival curve with the 95% confidence belts is constructed to visualize the survival experience of the patients. The curve demonstrates a continuous decline in survival probability over time. The confidence intervals are relatively narrower at early times and become wider at later times.

The Weibull plot shows that the observed points lie close to a straight line, indicating that the Weibull distribution provides a good fit to the remission data.

Problem: Twenty patients were given a treatment and it was planned to terminate the treatment on 180 weeks. The survival times in weeks of 16 patients by 180 weeks are as follows.

70, 160, 105, 140, 20, 113, 121, 10, 44, 150, 60, 30, 30, 11, 11, 15.

It is assumed that the above data follows a two parameter Weibull distribution.

(i) Estimate the parameters using ML method.

(ii) Calculate the standard errors of the estimates. Construct the 95% confidence intervals of the parameters.

(iii) Estimate the 75th percentile and also estimate the survival probability after 200 weeks.

(iv) Comments on your findings.

Solution: R-code and output:

```
#(i)
```

```
# MLEs of Weibull distribution under Type-I censoring
```

```
x=c(70, 160, 105, 140, 20, 113, 121, 10, 44, 150, 60, 30, 30, 11, 11, 15)
```

```
quantile(x, 0.632)
```

```
63.2 %
```

```
86.8
```

```
n=20; t0=180; r=16
```

```
library(stats4) #loading package stats4 for mle()
```

```
logL=function(a,b){
  term1=-n*log(b)
  term2=b*n*log(a)
  term3=-(b-1)*sum(log(x))
  term4=sum((x/a)^b)
  term5=(n-n)*((10/a)^b)
  terms=term1+term2+term3+term4+term5
}
```

```
coefs=coef(mle(minuslogl=logL, start=list(a=quantile(x,
0.632), b=1.5)))
```

```
coefs
```

a	b
113.139802	1.006442

```
##(i)
```

```
var=vcov(mle(minuslogl=logL, start=list(a=quantile(x,
0.632), b=1.5)))
```

```
var
```

	a.63.2 %	b
a.63.2 %	792.4752506	0.3827777
b	0.3827777	0.04615769

```
diag(var)
```

	a.63.2 %	b
a.63.2 %	792.4752506	0.04615769

```
sqrt(diag(var))
```

	a.63.2 %	b
a.63.2 %	28.1509369	0.2148434

95 % confidence interval for scale parameter

$$LCL = \text{coefs}[1] - 1.96 * \text{sqrt}(\text{diag}(\text{var})[1])$$

LCL

a

57.96397

$$UCL = \text{coefs}[1] + 1.96 * \text{sqrt}(\text{diag}(\text{var})[1])$$

UCL

a

168.3156

95 % confidence interval for shape parameter

$$LCL = \text{coefs}[2] - 1.96 * \text{sqrt}(\text{diag}(\text{var})[2])$$

LCL

b

0.5853491

$$UCL = \text{coefs}[2] + 1.96 * \text{sqrt}(\text{diag}(\text{var})[2])$$

UCL

b

1.427535

(iii)

Estimated the 75th percentile and survival probability
after time 200 weeks.

$$\text{qweibull}(0.75, \text{shape} = \text{coefs}[2], \text{scale} = \text{coefs}[1])$$

156.5175

$$1 - \text{pweibull}(200, \text{shape} = \text{coefs}[2], \text{scale} = \text{coefs}[1])$$

0.169615

(iv) Comments on my findings:

The maximum likelihood estimates of the Weibull parameters are scale parameter $\hat{\alpha} = 113.14$ and shape parameter $\hat{\beta} = 1.0064$. Since the shape parameter is very close to 1, the failure rate is approximately constant over time, so the Weibull distribution is close to an exponential distribution.

The standard errors are 28.15 for the scale parameter and 0.2148 for the shape parameter. The 95% confidence intervals are (57.96, 168.32) for the scale parameter and (0.5853, 1.4275) for the shape parameter, showing some uncertainty in the estimates.

The estimated 75th percentile is 156.52 weeks, meaning about 75% of failures are expected to occur by this time. The estimated survival probability after 200 weeks is 0.1696, meaning only about 16.96% of individuals are expected to survive beyond 200 weeks.

Problem: Standard and test preparations of an agent are tested at the same three dose levels (0.25, 0.50, and 1.00 mg/cc); and there are 8 replications at each dose of each preparation. The amounts of decrease in growth in millimeter (mm) of bacterial cultures are given in the following table.

Dose (mg/cc)	Preparation					
	Standard Preparation			Test Preparation		
	0.25	0.50	1.00	0.25	0.50	1.00
	6.0	9.4	12.8	4.9	8.2	11.0
	6.8	8.8	13.6	4.8	8.1	11.5
	6.2	9.4	13.4	4.9	8.1	11.4
Response	6.6	9.6	13.8	4.8	8.2	11.8
(mm)	6.4	9.8	12.8	5.3	7.6	11.8
	6.0	9.2	14.0	5.1	8.3	11.4
	6.9	10.8	13.2	4.9	8.2	11.7
	6.3	10.6	12.8	4.7	8.1	11.4

- (i) Assuming a parallel line assay system, estimate the "relative potency" P of the test preparation with its standard error.
- (ii) Construct a 95% confidence interval for P .
- (iii) Assuming a slope-ratio assay system, estimate the "relative potency" P of the test preparation with its standard error.
- (iv) Construct a 95% confidence interval for P .
- (v) Comment on your findings.

Solution: R-code and output:

```
#(i)
```

```
x=c(rep(1, 24), rep(0, 24), rep(0, 24), rep(1, 24), rep(log(0.25), 8), rep(log(0.5), 8), rep(log(1), 8), rep(log(0.25), 8), rep(log(0.5), 8), rep(log(1), 8))
```

```
x=matrix(x, 48, 3)
```

```
x
```

```
ys1=c(6, 6.8, 6.2, 6.6, 6.4, 6.0, 6.9, 6.3)
```

```
ys2=c(9.4, 8.8, 9.4, 9.6, 9.8, 9.2, 10.8, 10.6)
```

```
ys3=c(12.8, 13.6, 13.4, 13.8, 12.8, 14, 13.2, 12.8)
```

```
yt1=c(4.9, 4.8, 4.9, 4.8, 5.3, 5.1, 4.9, 4.7)
```

```
yt2=c(8.2, 8.1, 8.1, 8.2, 7.6, 8.3, 8.2, 8.1)
```

```
yt3=c(11, 11.5, 11.4, 11.8, 11.8, 11.4, 11.7, 11.4)
```

```
y=c(ys1, ys2, ys3, yt1, yt2, yt3)
```

```
y
```

```
b=solve(t(x)%*%x)%*%t(x)%*%y
```

```
b
```

```
[,1]
```

```
[1,] 13.168750
```

```
[2,] 11.543750
```

```
[3,] 4.800079
```

```
b[2]
```

```
[2] 13.16875
```

```
M=(b[2]-b[1])/b[3]
```

```
M
```

```
[1] -0.3343567
```

```
yhat=x%*%b
```

yhat

s = sum (y - yhat)^2 / 45 # Estimated log-potency

s

[1] 0.1594167

m1 = exp(M) # Estimated of potency

m1

[1] 0.7157984

v = s * solve(f(x) %*% x)

v

h = (1/L[3]) * c(-1, 1, -M)

h

var.M = t(h) %*% v %*% h

var.M

s.e.M = sqrt(var.M)

s.e.M ; s.e.m1 = sqrt(m1^2 * var.M) ; s.e.m1

[1]

[1.] 0.01770067

##(ii)

LCL = exp(M - 1.96 * s.e.M)

LCL

[1]

[1.] 0.6819324

UCL = exp(M + 1.96 * s.e.M)

UCL

[1]

[1.] 0.7513462

#(iii)

$x = c(\text{rep}(1, 48), \text{rep}(0.25, 8), \text{rep}(0.5, 8), \text{rep}(1, 8), \text{rep}(0, 24),$
 $\text{rep}(0, 24), \text{rep}(0.25, 8), \text{rep}(0.5, 8), \text{rep}(1, 8))$

$x = \text{matrix}(x, 48, 3)$

x

y

$b = \text{solve}(t(x) \% * \% x) \% * \% t(x) \% * \% y$

b

$\pi_2 = b[3] / b[2]$

π_2

$[1] 0.7699201$

$yhat = x \% * \% b$

$yhat$

$s = \text{sum}((y - yhat)^2) / 45$

s

$[1] 0.4873862$

$v = \text{solve}(t(x) \% * \% x)$

v

$s.e.\pi_2 = \text{sqrt}(t(h) \% * \% v \% * \% h)$

$s.e.\pi_2$

$[1]$

$[1,] 0.02845116$

#(iv) $LCL = \pi_2 - 1.96 * s.e.\pi_2 ; LCL$

$[1,] 0.7141558$

$UCL = \pi_2 + 1.96 * s.e.\pi_2 ; UCL$

$[1,] 0.8256844$

(v) Comment on findings:

The estimated relative potency of the test preparation is 0.7158 under the parallel line assay and 0.7699 under the slope-ratio assay. Since both estimates are less than 1, the test preparation appears to be less potent than the standard preparation.

The standard error of the relative potency is 0.0177 for the parallel line assay and 0.0285 for the slope-ratio assay. These relatively small standard errors indicate that potency estimates are reasonably precise.

The 95% confidence intervals are (0.6819, 0.7513) for the parallel line assay and (0.7142, 0.8257) for the slope-ratio assay. Since both confidence intervals lie entirely below 1, there is evidence that the test preparation has lower potency than the standard preparation.

Problem: The following table shows the proportions of infant mortality by birthweight for 35840 live births in a community.

Birthweight	Outcome at One Year		
	Dead	Alive	Total
≤ 2.5 kg	0.0140	0.1147	0.1287
> 2.5 kg	0.0088	0.8625	0.8713
Total	0.0228	0.9772	1

- (i) Estimate the odds ratio and comment on the result.
- (ii) Test the independence of birthweight and infant mortality using odds ratio.
- (iii) Estimate the attributable risk with its 95% confidence interval and comment on the result.

Solution: (i) and (ii): We set up a hypothesis,

H_0 : There is no relationship between birthweight and infant mortality.

H_1 : H_0 is not true

R-code and output:

$n = 35840$; $p_{11} = 0.0140$, $p_{12} = 0.1147$; $p_{21} = 0.0088$, $p_{22} = 0.8625$

$n_{11} = n * p_{11}$; $n_{12} = n * p_{12}$; $n_{21} = n * p_{21}$;

$n_{22} = n * p_{22}$

```
data = matrix(c(n11, n12, n21, n22), nrow = 2, byrow = TRUE,
  dimnames = list("Birthweight" = c("<= 2.5 kg", "> 2.5
  kg"), "Outcome" = c("Dead", "Alive")))
```

```
data
```

```
library(epitools)
```

```
odds.ratio = oddsratio(data)
```

```
odds.ratio
```

```
OR = (n11 * n22) / (n12 * n21)
```

```
OR
```

```
[1] 11.96303
```

Comment: OR = 11.96 and p-value < 0.05, indicating that we may reject our null hypothesis. Therefore, there is a strong relationship between birthweight and infant mortality. The odds of low birthweight yield infant mortality 11.96 times those for high birthweight.

```
#(iii)
```

```
ar = (p11 * p22 - p12 * p21) / ((p11 + p21) * (p21 + p22))
```

```
ar
```

```
[1] 0.5570241
```

```
#95% CI
```

```
se = sqrt((p12 + ar * (p11 + p22)) / (n * p21))
```

```
se
```

[1] 0.04372287

$$LCL = 1 - \exp(\log(1 - ar) - 1.96 * se)$$

LCL

[1] 0.5173884

$$UCL = 1 - \exp(\log(1 - ar) + 1.96 * se)$$

UCL

[1] 0.5934046

Comment: The value of attributable risk is 0.6570, indicates that 55.70% infant mortality would be prevented if low birthweight among infant could have been eliminated.

The 95% confidence interval is (0.51, 0.59). That is, we are 95% confident that the true attributable risk lies between 0.51 and 0.59.